COURSE NAME

DISCRETE MATHEMATICS

CSC 1204

(UNDERGRADUATE)

CHAPTER 12

PIGEONHOLE PRINCIPLE

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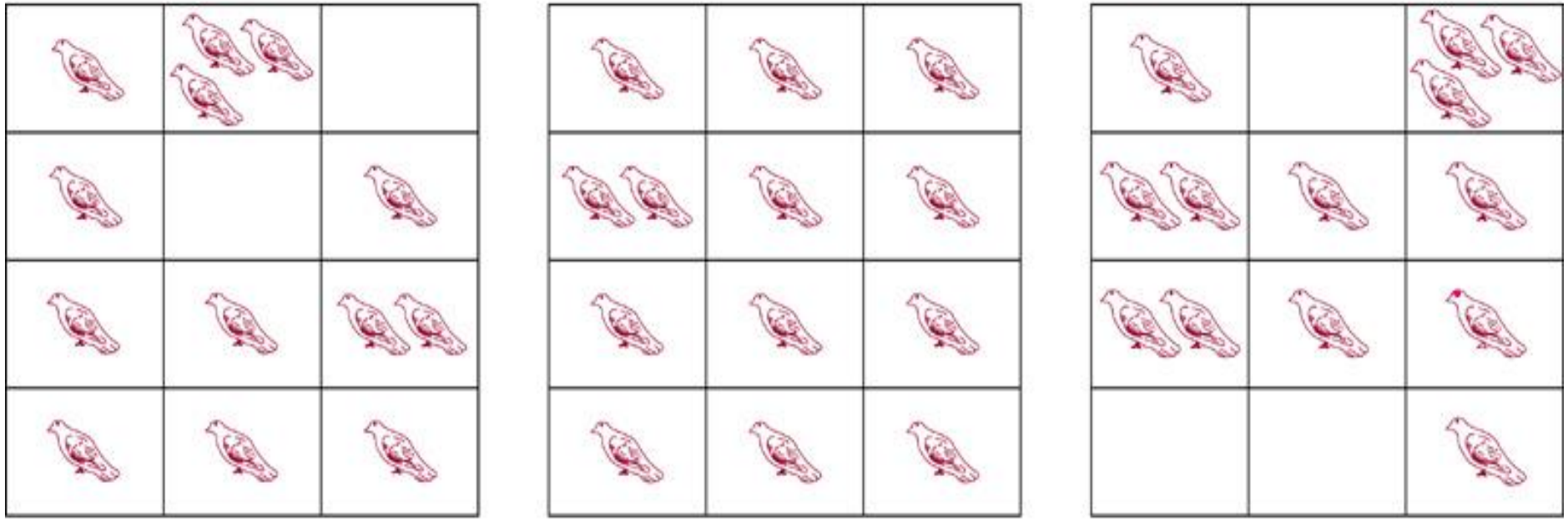




THE PIGEONHOLE PRINCIPLE

- If there are more pigeons than pigeonholes, then there must be at least one pigeonhole with at least two pigeons in it. Also called the Dirichlet drawer principle
- Theorem 1: (The Pigeonhole Principle) If k is a positive integer and k+1 or more objects are placed into k boxes, then there is at least one box containing two or more of the objects.

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(a) (b) (c)

THE PIGEONHOLE PRINCIPLE

- The Pigeonhole Principle can be used to prove a useful corollary about functions.
- Corollary 1: A function from a set with k+1 or more elements to a set with k elements is not one-to-one.
- Example: Among any group of 367 people, there must be at least two with the same birthday, because there are only 366 possible birthdays.
- Theorem 2 (The Generalized Pigeonhole Principle): If N objects are placed into k boxes, then there is at least one box containing at least $\lceil N/k \rceil$ objects.
- Example: Among 100 people there are at least $\lceil 100/12 \rceil = 9$ who were born in same month.

THE PIGEONHOLE PRINCIPLE

Q. What is the minimum number of students required in a discrete mathematics class to be sure that at least six will receive the same grade, if there are five possible grades, A, B, C, D, and F?

Solution:

- The minimum number of students needed to ensure that at least six students receive the same grade is the smallest integer N such that $\lceil N/5 \rceil = 6$.

  $\lceil 25 / 5 \rceil = 5, \quad \lceil 26 / 5 \rceil = \lceil 5.2 \rceil = 6$

  $N_{\text{smallest}} = 26$

- *Note:* The smallest such integer is $N = 5x5 + 1 = 26$. [where $5x6 = 30$]

  $\lceil N/5 \rceil \geq 6$, or, $N \geq 5x5 + 1$, or, $N \geq 26$

THE PIGEONHOLE PRINCIPLE

Q: There are 38 different time periods during which classes at a university can be scheduled. If there are 677 different classes, how many different rooms will be needed?

Solution:

- Thinks that the 38 time periods are the pigeonholes, and the 677 classes are the pigeons.
- By the generalized pigeonhole principle there is some time period in which at least

  $\lceil 677/38 \rceil = 18$ classes are meeting.

- Since each class must meet in a different room, we need 18 rooms.

REFERENCES

- *Discrete Mathematics and its applications with combinatorics and graph theory (7th edition)* by Kenneth H. Rosen [Indian Adaptation by KAMALA KRITHIVASAN], published by McGraw-Hill, is the required text.
- *SCHAUM'S ouTlines Discrete Mathematics(2nd edition)*, by Seymour Lipschutz, Marc Lipson
- *Discrete and Combinatorial Mathematics(3rd Edition)* by *R. P. Grimaldi,*, published by Addison-Wesley